PREFACE

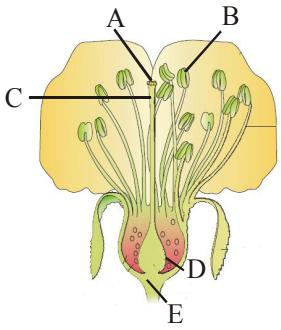
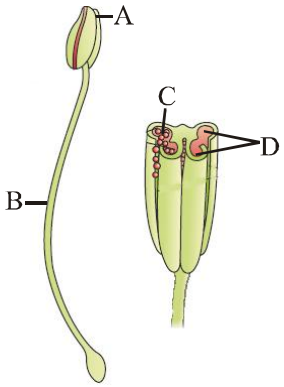
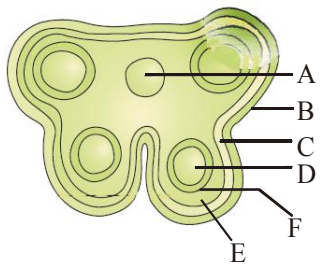
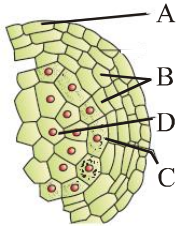
Prime aim of this NCR is to provide adequate knowledge with perfect revision of NCERT. Every chapter is provided with different types of questions based on NCERT to cover the entire content of each chapter. The questions provided are such that they help in achieving a thorough grasp of subject in a very short time for NEET. The answers are provided on the same page against the questions in most appropriate manner for their medical entrance preparation. When you read NCR, hide right side (Answers) by your palm, which will help you to memorise the NCERT rapidly and in very short time.

We wish our students make the best use of this NCR to get success.

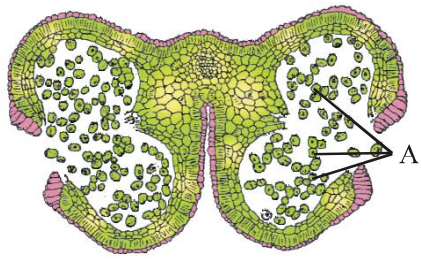
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

Serial No.	CONTENTS	Page No.
1.	Sexual reproduction in flowering plants	1
2.	Human reproduction	8
3.	Reproductive health	13
4.	Genetics (Principles of inheritance and variation)	14
5.	Genetics (Molecular basis of inheritance)	18
6.	Evolution	22
7.	Human health and disease	25
8.	Strategies for enhancement in food production	28
9.	Microbes in human welfare	33
10.	Biotechnology : Principles & processes	35
11.	Biotechnology and its application	39
12.	Organisms and Populations	41
13.	Ecosystem	46
14.	Biodiversity and Conservation	47
15.	Environmental Issues	49

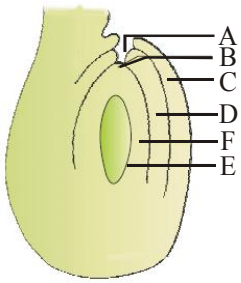
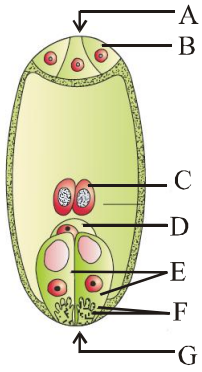
SEXUAL REPRODUCTION IN FLOWERING PLANTS

Q.NO.	QUESTION	ANSWER
1.	 Write down the name of A, B, C, D and E in given figure.	A-Stigma, B-Anther, C-Style, D-Ovary, E-Thalamus
2.	 Write down the name of A,B,C and D in given figure.	A-Anther, B-Filament, C-Pollen grain, D-Pollen sac
3.	 Write down the name of parts A,B,C,D,E and F in given figure.	A-Connective, B-Epidermis, C-Endothecium, D-Sporogenous tissue E-Middle layer, F-Tapetum
4.	 Write down the name of parts A,B,C, and D in given figure.	A-Epidermis, B-middle layer, C-Tapetum, D-microspore mother cell

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\01-SEXUAL REPRODUCTION.P65

5.	 Write the name of labelled structures 'A'	A-Pollen grains
	FILL IN THE BLANKS	
6.	The microsporangium develops further and becomes _____	Pollen sac
7.	The innermost wall layer of microsporangium is the ____	Tapetum
8.	In the centre of each microsporangium a compactly arragned tissue is _____	Sporogenous tissue
9.	There are the two part of a typical ___, the long and slender stalk is called __ and the terminal generally bilobed structure is called ___.	Stamen, Filament, anther
10.	A typical angiosperm anther is ____ with each lobe having two theca i.e. they are called ____	Bilobed, dithecous
11.	The hard outer layer of pollen grain, called exine is made up of __	Sporopollenin
	TRUE AND FALSE	
12.	In pollen grain exine has prominent apertures called germ pores where sporopollenin is absent.	True
13.	The inner wall of pollen grain is called intine.	True
14.	In microsporogenesis, PMC undergo mitosis and form pollen grains.	False
15.	Pollen grains can be stored in liquid nitrogen at –196°C.	True
16.	Wheat and rice pollen grains has pollen viability upto months.	False
17.	Match the following :- 1. Proximal part of filament — a. Tapetum 2. Inner most wall layers of pollen sac — b. attached to the thalamus 3. Sporopollenin — c. most resistant material 4. Ploidy of the cells of the tetrad — d. Haploid	1 - b, 2 - a, 3 - c, 4 - d
18.	In a mature pollen, how many nuclei are present and write their name also.	two nuclei, 1-vegetative nucleus 1-generative nucleus
19.	Name any plant that causes pollen allergy	*Parthenium*

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\01-SEXUAL REPRODUCTION.P65

20.	What is role of pollen tablets in athletes and race horses	to increase their performance
21.	Where is the use of pollen bank.	Crop breeding programmes
22.	What is the diameter of pollen grain	25-50 micrometer
23.	 In the given figure pistil, identify the labellings A,B,C & D.	A-Stigma, B-Style C-Ovary, D-thalamus
24.	 Identify A,B,C,D and E in the given figure of ovule.	A - Micropyle, B - Micropylar end, C - Outer integument, D - Inner integument, E - Embryosac, F - Nucellus
25.	 Identify A,B,C,D,E,F and G	A-Chalazal end B-Antipodal cell C-Polar nuclei D-Egg cell E-Synergid cells F-Filiform apparatus G-Micropylar end
	FILL IN THE BLANKS	
26.	The basal part of the carpel is _____	Ovary
27.	_____ serves as a landing plateform for pollen grains	Stigma
28.	The mitotic divisions which occur during embryo sac formation from megaspore are _____.	Strictly free nuclear divisions

29. An ovule has generally one embryo sac. — True

30. Each ovule has one or two protective envelop, called exine and intine — False

31. Hilum is the region where the body of ovule fuses with funiculus — True

32. Nucellus is a mass of cells enclosed within integument. — True

33. Only the non-functional megaspore develops into female gametophyte — False

34. When a embryo sac develops from a single megaspore, it is called monosporic development — True

35. Match the following :- — 1-D, 2-A, 3-B, 4-C, 5-E

	Column-A		**Column-B**
1	Central cell	A	Egg apparatus
2	One egg cell and two synergids	B	Chalazal end
3	Antipodal cell	C	Meiosis
4	Megaspore mother cell	D	Polar nuclei
5	Female gametophyte	E	Embryosac

36. In flowring plants, female reproductive whorl is called — Gynoecium

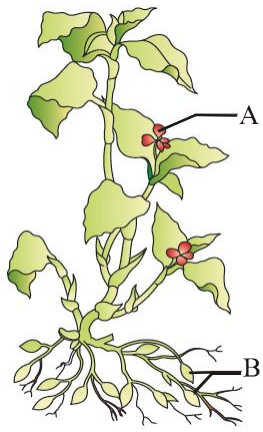
37. What we say if a gynoecium of flower has many carpels. — Multicarpellary

38. If a gynoecium of a flower has many carpels and they are fused, this condition is called — Multicarpellary, syncarpous

39. The process of formation of megaspores from the megaspore mother cell by meiosis, called — Megasporogenesis

40. In a typical angiosperm, mature embryosac is _____ celled. — 7/Seven

41.

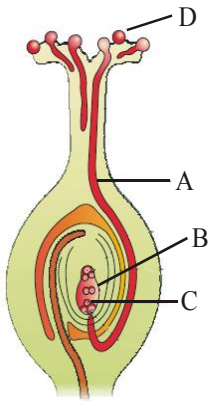
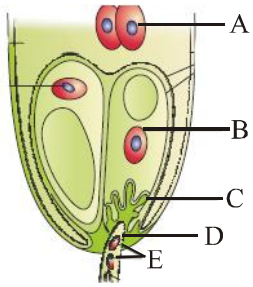
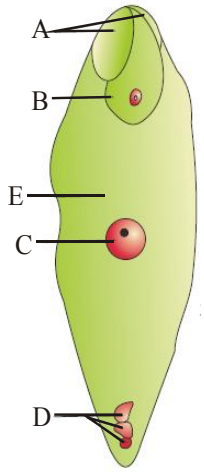


Identify A and B in this figure.

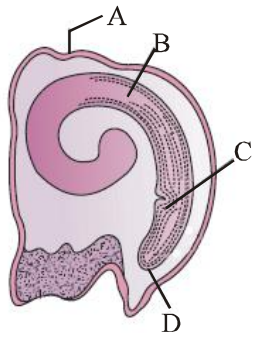
A- Chasmogamous flower
B- Cleistogamous flower

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\01-SEXUAL REPRODUCTION.P65

	FILL IN THE BLANKS	
42.	The transfer of pollen grains from ____ to ____ is known as ____	Anther, stigma, pollination
43.	Transfer of pollen grains from ____ to stigma of same flower is called ___	Anther, Autogamy
44.	For the formation of embryo sac the _____ undergoes one _____ division and three _____ divisions in megaspore nucleus.	Megaspore mother cell, Meiotic, Mitotic
45.	Transfer of pollen from anther to stigma of another flower of same plant is called ____	Geitonogamy
	TRUE/FALSE	
46.	Xenogamy is the only type of pollination in which genetically different pollens germinate on stigma	True
47.	Water is the most common pollinating agent in abiotic pollination	False
48.	Wind - pollination is quite common in grasses	True
49.	In *Vallisneria* the pollinating agent is water.	True
50.	Dioecious condition prevents autogamy and geitonogamy	True
51.	Match the following **Column-A** / **Column-B** 1 Out breeding device — A Water pollination 2 Chasmogamous and cleistogamous flower — B Prevents self pollination 3 Amorphophallus — C Oxalis 4 Zostera — D Tallest flower	1-B, 2-C, 3-D, 4-A
52.	Which genetic mechanism prevents the self-pollination	Self incompatibility
53.	Which thype of pollination is found in cleistogamous flower	Self pollination
54.	How many cells are present in a mature pollen grain	2-cells
55.	Which type of pollen grains germinate on the stigma	Compatible pollen grains.

56.	 Identify A,B,C and D in this figure.	A-Pollen tube, B-Embryosac C-Egg cell, D-Pollen grain
57.	 Identify A,B,C,D and E in this figure.	A-Polar nuclei, B-Synergid cell, C-Filiform apparatus, D-Pollen tube, E-male gametes
58.	 Identify A,B,C,D and E in this figure of fertilized embryosac.	A-Degenerating synergids B-Zygote C-Primary endosperm nucleus (PEN) D-Degenerating antipodal cells E-Primary endosperm cell (PEC)
	FILL IN THE BLANKS	
59.	Removal of anthers from the flower bud is called ____	Emasculation
60.	Two polar nuclei fuse together and form _____	Secondary nucleus
61.	First male gamete fuses with egg cell nucleus and form ___	Zygote
62.	Ploidy of the primary endosperm nucleus in typical angiosperms is _____	3n (Triploid)

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\01-SEXUAL REPRODUCTION.P65

63.	Endosperm is a nutritive tissue	True
64.	In coconut the endosperm is free-nuclear endosperm	True
65.	Ovary is a ripened ovule.	False
66.	The study of development of embryo is called embryology	True
67.	Two-cotyledons are found in monocot seeds.	False
68.	In mango, the mesocarp is edible part	True
69.	Seeds of pea and ground nut are non endospermic.	True
70.	Seeds of maize and wheat are endospermic.	True
71.	What is perisperm	residual, persistent nucellus in seed
72.	The term which is used for fruits which are developed without fertilization is _____	Parthenocarpic fruits
73.	 Identify A,B,C and D in figure of a seed.	A-Seed coat B-Cotyledon C-Shoot apical meristem D-Root tip
74.	Occurrence of more than one embryo in a seed is referred to as :-	Polyembryony

HUMAN REPRODUCTION

Q.NO.	QUESTION	ANSWER
1.	Male reproductive system is located in	Pelvic region
2.	Scrotum helps in maintaining low temperature of the testes about _____ lesser than body temperature.	2–2.5°C
3.	Each testis is _____ is shape.	Oval
4.	Length of each testis is __(A)__ in the length and __(B)__ in width.	A = 4 to 5 cm, B = 2 to 3 cm
5.	Each testis has how many lobules ?	250
6.	The_____ undergo meiotic divisions finally leading to sperm formation.	Germ cell
7.	Androgens are secreted by	Leydig cells
8.	Which is the correct pathway of sperms. (i) Epididymis (ii) Seminiferous tubules (iii) Vas-efferentia (iv) Rete-testis	ii – iv – iii – i
9.	Glans penis is covered by a loose fold of skin is called.	Fore skin
10.	The male accessory glands include paired __(A)__ a __(B)__ and paired __(C)__.	A-Seminal vesicle, B- prostate C- Bulbourethral
11.	Secretions of_____ glands also helps in the lubrication of penis.	Bulbourethral
12.	Primary spermatocyte is haploid.	False (Diploid)
13.	Secondary spermatocytes undergo Ist meiotic division	False (It show IInd meiotic division)
14.	Spermatids are transformed into spermatozoa by the process called.	Spermiogenesis
15.	Releasing of sperms from seminiferous tubules is known as.	Spermiation
16.	Spermatogenesis process start before birth.	False (At puberty due to availability of more GnRH)
17.	LH acts at the __A__ cells and stimulates synthesis of __B__.	A- Leydig, B-Androgens
18.	FSH acts on the __A__ cells and stimulates secretion of __B__ which help in the process of __C__.	A-Sertoli, B-Some factors, C-Spermiogenesis
19.	The sperm head contains an elongated_____ nucleus.	Haploid
20.	Acrosomal enzyme helps in.	Fertilization of ovum
21.	Which parts secretions are essential for maturation and motility of sperms.	Epididymis, Seminal vesicle, prostate
22.	Sperms + ? = Semen.	Seminal plasma
23.	Functions of male sex accessory ducts and glands are maintained by	Androgens
24.	Which of the following is not a part of female reproductive system? Ovary, oviduct, uterus, vasdeferens, hymen	Vasdeferens

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\02-HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH.P65

25.	Ovaries are the ..... female sex organs.	Primary
26.	Ovaries produce the (A) and several (B) hormones.	(A) female gamete (B) steroid
27.	Each ovary is about 5 to 8 cm in length in human. (True/false)	False
28.	The ovarian stroma is divided into two zones – a peripheral (A) and an inner (B) .	(A) Cortex (B) Medulla
29.	The oviduct, uterus and vagina constitute the female external genitalia. (True/false)	False
30.	Each fallopian tube is about ..... long.	10-12 cm
31.	Which help in collection of ovum after ovulation ?	Fimbriae
32.	The shape of the uterus is like an apple. (True/false)	False
33.	How many layers are present in uterus ? One/Two/Three/Four	Three
34.	Correct sequence of uterine wall outer to inner side ?	Perimetrium → myometrium → endometrium
35.	The (A) undergoes cyclic changes during menstrual cycle while the (B) exhibits strong contraction during delivery of the baby.	(A) Endometrium (B) Myometrium
36.	Which of the following does not include in external genitalia in female ? Mons pubis, labia majora, fimbriae, hymen	Fimbriae
37.	Match the column : (a) Mons pubis (i) paired folds of tissue (b) Labia majora (ii) tiny finger like structure (c) Clitoris (iii) cushion of fatty tissue (d) Labia minora (iv) fleshy folds of tissue	a-(iii), b-(iv), c-(ii), d-(i)
38.	The clitoris is a tiny finger like structure which lies at the upper. Junction of the (A) above the (B) .	(A) two labia minora (B) urethral opening
39.	The presence or absence of hymen is not a reliable indicator of virginity or sexual experience. (True/false)	True
40.	A functional mammary gland is characteristic of all animal female. (True/false)	False
41.	The glandular tissue of each breast is divided into ..... mammary lobes.	15-20
42.	The cells of ..... secrete milk.	Alveoli
43.	The alveoli open into	Mammary tubules
44.	Several mammary ducts join to form a wider (A) which is connected to (B) through which milk is sucked out.	(A) mammary ampulla (B) lactiferous duct

45.	The primary sex organs the (A) in the males and the (B) in the females.	(A) testis (B) ovaries
46.	Correct sequence of spermatogenesis : (a) spermatogonia (b) spermatids (c) primary spermatocyte (d) secondary spermatocyte (e) spermatozoa	a → c → d → b → e
47.	After spermiogenesis; sperm head become embedded into the ……	Sertoli cells
48.	Match the column : **Column-A** — **Column-B** (a) Spermiation — (i) Released of sperms from the seminiferous tubules (b) Insemination — (ii) Semen is released by the penis into the vagina (c) Ejaculation — (iii) Released of sperms outside the male body	a-(i), b-(ii), c-(iii)
49.	Match the column :- **Column-A** — **Column-B** (a) Spermatogonia — (i) multiply by mitotic division (b) Primary spermatocyte — (ii) undergo second meiotic division (c) Secondary spermatocytes — (iii) undergo meiosis-I (d) Spermatids — (iv) Transformed into sperms by the process of spermiogenesis	a-(i), b-(iii), c-(ii), d-(iv)
50.	How many structures are haploid ? Spermatogonia, spermatozoa, spermatids, primary spermatocyte, secondary spermatocytes	3 (Spermatozoa, secondary spermatocyte, spermatid)
51.	How many structures are diploid ? Sperms, spermatid, spermatogonia, primary spermatocyte, secondary spermatocytes	2 (Spermatogonia, primary spermatocyte)
52.	Spermatogenesis starts at the age of (A) due to significant increase in the secretion of (B).	(A) Puberty (B) GnRH
53.	(A) acts at the leyding cells and stimulates synthesis and secretion of (B).	(A) LH (B) androgens

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XI\ENGLISH\02-HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH.P65

54.	The sperm head contains an elongated diploid nucleus. (True/false)	False
55.	Which part of sperm produce energy for the movement of tail ?	Middle piece
56.	The human male ejaculates about _(A)_ million sperms during a coitus of which for normal fertility at least _(B)_ percent sperms must have normal shape and size and at least _(C)_ percent of them must show vigorous motility.	(A) 200-300 (B) 60 (C) 40
57.	The functions of male sex accessory ducts and glands are maintained by the	Testicular hormones (Androgens)
58.	The process of formation of a mature female gamete is called ?	Oogenesis
59.	The oogenesis is initiated during the ..... stage.	Embryonic development
60.	How many primary follicles are left in each ovary at puberty ?	60,000 – 80,000
61.	Correct sequence of follicular development : (a) Graafian follicle (Mature tertiary) (b) Primary follicle (c) Tertiary follicle (d) Secondary follicle	b $\rightarrow$ d $\rightarrow$ c $\rightarrow$ a
62.	The Graafian follicle ruptures to release the _(A)_ from the ovary by the process called _(B)_.	(A) Secondary oocyte (B) Ovulation
63.	The reproductive cycle in the female primates is known as	Menstrual cycle
64.	The first menstruation begins at _(A)_ and is called _(B)_.	(A) Puberty (B) Menarche
65.	In human females, menstruation is repeated at an average interval of about ?	28 or 29 days
66.	How many ovum is released during the middle of each menstrual cycle ?	One
67.	Menstrual phase is completed about ?	3-5 days
68.	The menstrual flow results due to breakdown of	Endometrium
69.	The menstrual phase is followed by the which phase ?	Follicular phase
70.	During follicular phase changes in the ovary and the uterus are induced by changes in the levels of _(A)_ and _(B)_ hormones.	(A) Pituitary (B) Ovarian
71.	Both LH and FSH attain a peak level about ..... of menstrual cycle.	mid - cycle
72.	Which hormone is mainly responsible for ovulation ?	LH
73.	Which hormone is essential for maintenance of endometrium ?	Progesterone
74.	In human beings, menstrual cycles ceases around the age of	50 years

75.	The process of fusion of a sperm with an ovum is called ?	Fertilisation
76.	After completion of second meiotic division of the secondary oocyte and results in the formation of a (A) and a haploid (B).	(A) Second polar body (B) Ovum
77.	How many chromosomes will be there in the zygote of human ?	46
78.	It is correct to say that the sex of the baby is determined by the father and not by the mother. (False/true)	True
79.	The (A) division starts as the zygete moves through the isthmus of the oviduct called (B).	(A) Mitotic (B) Cleavage
80.	The embryo with 8 to 16 blastomeres is called ?	Morula
81.	The blastomeres in the blastocyst are arranged into an outer layer called (A) and an inner group of cells attached to trophoblast called (B).	(A) Trophoblast (B) Inner cell mass
82.	Some finger like projection appear on the trophoblast called ?	Chorionic villi
83.	The structural and functional unit between developing embryo and maternal body called ?	Placenta
84.	The placenta is connected to the embryo through an (A).	Umbilical cord
85.	Placenta acts as an	Endocrine tissue
86.	Which of the following hormones are not secrete by placenta ? HCG, hPL, progesterone, oxytocin, estrogen, prolactin, relaxin	Oxytocin, prolactin
87.	Match the column :- **Column-A** / **Column-B** (a) After one month of pregnancy — (i) Heart is formed (b) End of second month of pregnancy — (ii) Major organ systems are formed (c) End of first trimester of pregnancy — (iii) First movement of foetus (d) During the fifth month of pregnancy — (iv) Limbs and digits develop	a-(i), b-(iv), c-(ii), d-(iii)
88.	Parturition is induced by a complex neuroendocrine mechanism. (True/false)	True
89.	The signals for parturition originate from the fully developed foetus only. (True/false)	False
90.	Which hormone acts on the uterine muscle and causes stronger uterine contractions.	Oxytocin
91.	The milk produced during the initial few days of lactation is called ?	Colostrum

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\02-HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH.P65

REPRODUCTIVE HEALTH

1.	Which country in world to initiate family planning programme in 1951	INDIA
2.	Improved programmes covering wider reproduction-related areas are currently in operation under the popular name.	'Reproductive and child Health care (RCH) programmes.
3.	Amniocentesis is statutory banned for	Sex determination
4.	Central Drug Research Institute (CDRI). Scientists developed a new oral contraceptive for the females is.	Saheli
5.	Periodic abstinence is which type of centraceptive method	Natural method
6.	In periodic abstinence couple avoid coitus from	10 to 17 day of m.c.
7.	In m.c. day 10 to 17 is known as	Fertile period
8.	Lactational amenorrhea is effective only upto maximum period of.	6 month
9.	'Nirodh' is a popular brand of	Condom for male
10.	Which contraceptive method also prevent from STD and AIDS.	Condom
11.	Match the columns.	a-v, b-iv, c-i, d-ii, e-iii, f-vi

a	Non-meclicated IUDs	i	Uterus unsuitable for implantation
b	Copper-releasing IUDs	ii	Inhibit ovulation
c	Hormone-releasing IUDs	iii	Cutting of fallopian tube
d	Progesterone-pill	iv	Suppress motility of sperms
e	Tubectomy	v	Increase phagocytosis of sperms
f	Vasectomy	vi	Cutting of vas-deferens

12.	Ideal contraceptives for the females who wants to delay pregnancy.	IUDs
13.	Oral pills for females causes	Inhibit ovulation and implantation
14.	Once a-week oral pill is	Saheli
15.	In emergency which contraceptive methods may be used within 72 hours of coitus.	IUD & progestogen or progestogen estrogen combination pills.
16.	Which contraceptive method are also known as sterilisation.	Surgical method
17.	Sterilisation in male is known as.	Vasectomy
18.	Government of INDIA legalised MTP in.	1971
19.	MTP is considered relatively safe during.	First trimester
20.	In ZIFT-early embryo is transfered in (upto 8 blastomeres)	Fallopian tube
21.	Full form of ZIFT	Zygote Intra Fallopian Transfer
22.	In IUT, embryos with more than 8 blastomeres are transffered in	Uterus
23.	Full form of GIFT	Gamete Intra Fallopian Transfer
24.	Procedure in which to form an embryo in the laboratory in which a sperm is directly injected into the cytoplasm of ovum	ICSI-Intracytoplasmic sperm injection
25.	If male is unable to inseminate the semen into vegina then which procedure is used.	Artificial insemination (AI) or IUI-Intrauterine insemination.

GENETICS
(PRINCIPLES OF INHERITANCE AND VARIATION)

Q.NO.	QUESTION	ANSWER
1.	Dominance is not an autonomous feature of a gene in case more more than one phenotype is influences by the same gene.	TRUE
2.	Mendelian recombinants are due to crossing over.	FALSE
3.	Mendel Conducted Somatic Hybridization experiments on garden pea.	FALSE
4.	Two alleles of a gene are located on homologous sites on non homologous chromosomes.	FALSE
5.	Mutation is a phenomenon which results in alteration of DNA sequences.	TRUE
6.	Drosophila complete their life cycle in four weeks.	FALSE
7.	In Drosophila single mating could produce large number of progeny flies.	TRUE
8.	Drosophila has many types of hereditary variations that can be seen with low power microscopes.	TRUE
9.	A true breeding line is one that, having undergone continuous self pollination, shows the stable trait inheritance and expression for several generations.	TRUE
10.	Mendel selected seven visible characters each with two contrasting traits.	TRUE
11.	The gene which affect more than one phenotype called pleiotropic gene.	TRUE
12.	In a test cross. F_1 hybrids are crossed with homozygous dominant plants.	FALSE
13.	Inheritance of one character is always independent to the inheritance of other character.	FALSE
14.	Pisum sativum reproduces asexually in nature.	FALSE
15.	Emasculation is the process of removal of pistils from unisexual flowers.	FALSE
16.	Sickle cell anaemia follows the same inheritance pattern as that haemophilia.	FALSE
17.	Mendel experiments had a large sampling size.	TRUE

18.	In a dissimilar pair of factors one member of the pair dominates the other.	TRUE
19.	Law of segregation is based on the fact that the alleles do not show any blending	TRUE
20.	Multiple alleles can be found only when population studies are made.	TRUE
21.	Mendel widely publicised his work.	FALSE
22.	Mendel's approach of using mathematics to explain biological phenomena was totally new.	TRUE
23.	Mendel could not provide any physical proof for the existence of factor.	TRUE
24.	Chromosomes as well as genes occurs in pairs.	TRUE
25.	Gene for same character segregate at the time of	*Gamete formation*
26.	Genes and chromosomes are found in pairs in _____ cells.	Diploid
27.	Alleles are slightly different forms of the same.....	Gene
28.	When modified allele is equivalent to the unmodified allele it will produce.....	Same phenotype
29.	Drosophila complete their life cycle in about..... weeks	two
30.	Grass hopper is an example of ..... type of sex detamination	XO
31.	..... disease shows its transmission from unaffected carrier female to some of the male progeny	Sex linked recessive
32.	The possibility of a female becoming a haemophilic is extremely rare because mother of such a female has to be at least carrier and the father should be.....	haemophilic
33.	In a monohybrid cross, we study ..... character.	One
34.	Holandric genes are present on	Y-chromosome
35.	Behaviour of Chromosomes was parellel to the behaviour of genes explained by	Sutton and Boveri
36.	During spermatogenesis in human males, ... type of gametes are produced.	two
37.	The gene, which affects many phenotypes.	Pleiotropic gene.
38.	Mendelian recombinants are due to	Independent assortment
39.	Genes present on same chromosome are inherited together called :-	Linked gene
40.	Characters are controlled by discrete units called :-	Factors

41.	In which case F_1 generation resembled eithes of the two parents?	Dominance
42.	In human beings 'ABO' Blood group is controlled by which gene?	Gene I
43.	How many phenotypes are possible in human 'ABO' Blood group?	Four
44.	Chromosomal theory of inheritence was proposed by.	Sutton & Boveri
45.	All genes on single chromosome forms :-	Linkage group
46.	Holandric genes are always transfer from father to	Son
47.	Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the	Chromosomal theory of Inheritance
48.	The unmodified allele, which represents the original phenotype is the dominant allele and the modified allele is generally....	Recessive
49.	Recessive trait is seen due to	Non functional enzyme or no enzyme.
50.	Match the columns :-	A-3, B-1, C-4, D-2

	Column-I	**Column-II**
A	Monohybrid cross	1.Codominance
B	'AB' Blood group	2. Correns
C	Dihybrid Cross	3. Dominance
D	Mirabilis jalapa	4. Independent assortment

51.	Match the columns	A-2, B-3, C-1

	Column-I	**Column-II**
A	Hypertrichosis	1. Autosomal Recessive
B	Colourblindness	2. Y-chromosome
C	Sickle cell Anaemia	3. X- Linked

52.	Match the column	1- (E),2- (A),3- (B),4- (C), 5- (D)

	Column-I	**Column-II**
1	Sickle cell anaemia	(A) Haemophilia
2	Bleeder's disease	(B) Hugodevries
3	Mutation	(C) 'O' Blood group
4	Universal donor	(D) Mendel
5	Pisum sativum	(E) Sickle shape RBCs

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\03-GENETICS.P65

53.

I	II
(A) 	(i) Death
(B) 	(ii) Five unaffected offspring
(C) 	(iii) Sex unspecified
(D) 	(iv) Female
(E) 	(v) male

(A) -v, (B) -iv, (C) -iii, (D) -ii,(E) -i

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\03-GENETICS.P65

GENETICS
(MOLECULAR BASIS OF INHERITANCE)

Q.NO.	QUESTION	ANSWER
1.	In living world, most abundant genetic material is	DNA
2.	Which end of a polynucleotide chain has a free phosphate moiety ?	5'–end
3.	5-methyl uracil is another chemical name for	Thymine
4.	The backbone in a polynucleotide chain is formed due to	Sugars and phosphates
5.	What generates approximately uniform distance between the two strands of DNA double helix	Purine–pyrimidine hydrogen bonding
6.	Flow of information from RNA to DNA is called	Reverse transcription
7.	Histones are rich in which basic amino acids ?	Lysine and Arginine
8.	A segment of DNA coding for a polypeptide is called as	Cistron
9.	The RNA polymerase is only capable of catalysing the process of	Elongation
10.	In eukaryotes, the hnRNA is transcribed by	RNA polymerase II
11.	Which scientist suggested the code should be made up of three nucleotides ?	Geoge gamow
12.	In their experiments, Hershey and Chase used radioative _____ labelled DNA and radioactive _____ labelled protein capsule.	^{32}P, ^{35}S
13.	The genetic material must have four qualities, These are :	1. Replicability 2. Stability 3. Mutability 4. Expressibility
14.	Which centrifugation technique was used by Meselson and Stahl in their experiment which proved semi-conservative DNA replication?	CsCl - density gradient Centrifugation
15.	Most Prokaryotic genes are	Polycistronic and Non-split
16.	Most eukaryotic genes are	Monocistronic and Split
17.	Which scientist formed co-polymer of nucleotides	H.G. Khurana
18.	Some amino acids are coded by more than one codon, hence the code is	Degenerate
19.	A classical example of point mutation is	Sickle cell anaemia
20.	Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion. Such mutations are referred to as	Frame-shift mutations
21.	The t-RNA is a compact molecule which in 3-D looks like	Inverted L

22.	An m-RNA also have some additional sequences that are not translated and are referred to as	Untranslated regions (UTR)
23.	In eukaryotes, the regulation of gene expression could be exerted at	Four levels
24.	In most cases the sequence of the operator is bound by	Repressor protein
25.	Each operon has its specific operator and specific	Repressor
26.	The operon concept was given by	Jacob and Monod
27.	the i gene of lac operon codes for	Repressor
28.	The a gene of lac operon codes for	Transacetylase enzyme
29.	In lac operon, lactose acts as	Inducer
30.	Regulation of lac operon by repressor is referred to as	Negative regulation
31.	The repressor is inactivated by interaction with the	Inducer
32.	The fragments were sequenced using automated DNA sequencers that worked on the principle of a method developed by	F. Sanger
33.	DNA finger printing involves identifying differences in some specific regions in DNA sequence called as	Repetitive DNA
34.	The bulk DNA forms major peak and the other small peaks are referred to as	Satellite DNA
35.	If inheritable mutation is observed in a population at high frequency, it is referred to as	DNA polymorphism
36.	The technique of DNA fingerprinting was initially developed by	Alec Jeffreys
37.	VNTR belongs to a class of satellite DNA referred to as	Mini-satellite
38.	The sensitivity of DNA fingerprinting technique has been increased by the use of	Polymerase chain reaction
39.	HGP was closely associated with the rapid devlopment of new area in biology called	Bioinformatics
40.	**Fill in the blank** The process of translation requires transfer of genetic information from polymer of _____ to polymer of amino acids.	Nucleotides
41.	Marshall Nirenberg's _____ for protein synthesis finally helped the code to be deciphered	Cell free system
42.	AAA $\longrightarrow$ A CCC $\longrightarrow$ B GGG $\longrightarrow$ C UUU $\longrightarrow$ D AUG $\longrightarrow$ E UAA, UAG, UGA $\longrightarrow$ F	A = Lys B = Pro C = Gly D = Phe E = Met and Start codon F = Stop codon

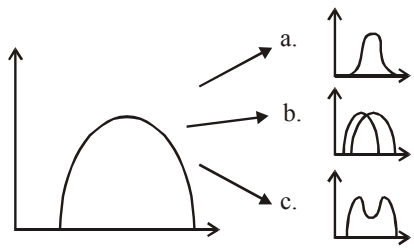
43.	Translation refers to process of polymerization of _____ to form a polypeptide.	Amino acids
44.	The ribosome also act as _____ for the formation of peptide bond	Catalyst
45.	A translational unit in __A__ is the sequence of RNA that is flanked by the __B__ and the __C__ and codes for a polypeptide	A = m-RNA, B = Start codon, C = Stop codon
46.	The UTRs are present at both __A__ and __B__ (After stop codon)	A = 5' end, B = 3' end
47.	At the end, a _____ binds to stop codon, terminating translation and releasing the complete polypeptide from the ribosome	Release factor
48.	In prokaryotes, control of the rate of _____ is predominant site for control of gene expression	Transcriptional initiation
49.	The activity of RNA polymerase at a given promoter is in turn regulated by interaction with _____, which affect its ability to recognise start site	Accessory protein (Regulatory protein)
50.	The (z) gene codes for __A__, which is primarily responsible for hydrolysis of _____.	A = β-glactosidase B = lactose
51.	The (y) gene codes for __A__, which increases permeability of cell to __B__.	A = Permease B = β-galactoside (Lactose)
52.	Human genome project was launched in	1990
53.	The commonly used hosts for cloning in HGP were __A__ and __B__ and the vectors were called as __C__ and __D__ .	A = Bacteria, B = Yeast, C = BAC, D = YAC
54.	The human genome contains _____ million nucleotide bases	3164.7
55.	Less than _____ of the human genome codes for proteins	2 percent
56.	Chromosome 1 has most genes __A__ and y has the fewest __B__.	A = 2968, B = 231
57.	A typical nucleosome contains _____ bp of DNA	200
58.	The nucleosomes in chromatin are seen as _____ structure when viewed under electron microscope	'Beads-on-string'
59.	Transcriptionally active chromatin is called as _____.	Euchromatin
60.	The biochemical nature of 'transforming principle' in Griffith's experiment was determined by _____ .	Avery, MacLeod and McCarty
61.	The unequivocal proof that DNA is the genetic material came from the experiments of _____ .	Alfred Hershey & Martha Chase
62.	The first genetic material was _____ .	RNA
63.	The DNA in chromosomes also replicates semiconservatively was proved by _____ through his experiments performed on *Vicia faba* using radioactive _____ .	Taylor, thymidine
64.	In *E.coli*, the average rate of polymerisation during DNA replication is approximately _____ .	2000 bp per second
65.	In a transcription unit, the promoter is said to be located towards _____ of the structural gene.	5'–end (upstream)

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\03-GENETICS.P65

66.	During DNA replication, leading strand is synthesized _____ while lagging strand is synthesized _____ .	Continuously, Discontinuously
67.	A piece of DNA if needed to be propagated during recombinant DNA procedures requires a vector because it provides_____.	Origin of replication
68.	In bacteria, σ-factor helps in transcription _____ whereas ρ-factor helps in transcription _____.	Initiation, termination
	True/False	
69.	No complementarity exists between nucleotides and amino acids	True
70.	Severo Ochoa enzyme (Polynucleotide phosphorylase) was also helpful in polymerising RNA with defined sequence in a template dependent manner.	False
71.	The code is nearly universal	True
72.	The codon is read in m-RNA in contiguous fashion	True
73.	There is one t-RNA for stop codons	False
74.	Formation of peptide bond requires energy	True
75.	First phase of translation is charging of t-RNA or aminoacylation of t-RNA	True
76.	For initiation, the ribosome binds to the m-RNA at the start codon that is recognised only by the initiator t-RNA	True
77.	The development and differentation of embryo into adult organism are also a result of the coordinated regulation of expression of several sets of genes.	True
78.	Regulatory proteins can act only positively	False
79.	In Lac operon, three structural genes are regulated by a common promoter and regulator gene	True
80.	The genes present in the operon are needed together to function in same or related metabolic pathway	True
81.	A very low level of expression of lac operon has to be present in cell all the time	True
82.	The repressor of the lac operon is synthesized all the time	True
83.	Lac operon is under control of positive regulation as well	True
84.	Total number of genes in human genome is approximately 30,000	True
85.	The split-gene arrangements represent probably a modern feature of the genome.	False
86.	The presence of introns is reminiscent of antiquity.	True
87.	Process of splicing represents the dominance of RNA world.	True
88.	The process of transription and translation can be coupled in bacteria.	True
89.	In DNA replication, deoxyribonucleoside triphosphates serve dual purposes as _____	Substrate and energy source

EVOLUTION

Q.NO.	QUESTION	ANSWER
1.	The study of history of life forms on earth is known as :	Evolutionary Biology
2.	Stellar distances are measured in.	Light years
3.	The universe is very old almost.	20 Billion years
4.	Which theory attempts to explain to us the origin of universe?	Big-Bang Theory
5.	In the solar system of the milky way galaxy, the earth was supposed to have been formed about :	4.5 Billion years back
6.	Life come out of decaying and rotting matter like straw, mud etc. This was the theory of :	Spontaneous generation
7.	The scientist who by careful experimentation demostrated that life comes only from pre-existing life :	Louis Pasteur
8.	Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules (eg. RNA, protein etc) and that formation of life was preceded by :	Chemical evolution
9.	In 1953, S.L. Miller, an American scientist, created electric discharge in a closed flask containing which gases at 800°C.	CH_4, H_2, NH_3 and water vapour
10.	Based on observations made during a sea voyage in a sail ship called H.M.S. Beagle round the world who concluded that existing living forms share similarities :	Charles-Darwin
11.	The fitness, according to Darwin, refers ultimately and only to	Reproductive fitness
12.	According to Darwin, mechanism of evolution is	Natural selection
13.	Remains of hard parts of life form found in rocks are	Fossils
14.	Study of fossils is known as :	Paleontology
15.	The same structure developed along different directions due to adaptions to different needs. These structures are called as :	Homologous structure
16.	Analogous structures are a result of :	Convergent evolution
17.	The flippers of Penguins and Dolphins are an example of :	Analogy
18.	Which can be used as industrial pollution indicators?	Lichens
19.	Moths that were able to _____ themselves. i,e hide in the background, survived.	Camouflage
20.	We employ antibiotics or drugs against eukaryotic organisms/cells. Hence, resistant organisms/cells are appearing in a time scale of month or years and not centuries. These are example of evolution due to	Anthropogenic-action
21.	On which island Darwin observed many varieties of finches?	Galapagos Islands
22.	Who worked in Malay Archipelago?	Alfred Wallace

23.	The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called :	Adaptive radiation
24.	Two examples of adaptive radiations are :	Darwin's finches and Australian-marsupials
25.	When more than one adaptive radiations appeared to have occured in an isolated geographical area (representing different habitats). One can call this :	Convergent evolution
26.	The rate of appearance of new forms is linked to the life cycle or the life span. True/False	True
27.	The geological history of earth closely correlates with the biological history of earth. True/False	True
28.	Acording to Darwin, adaptive ability is acquired. True/False	False (It is inherited)
29.	Two key concepts of Darwinian theory of evolution are :	Branching descent and natural selection
30.	The work of Thomas Malthus on populations influenced :	Darwin
31.	Hugo de Vries based on his work on which plant brought forth the idea of mutation?	Evening-primrose
32.	Single step large mutation is called :	Saltation
33.	Which principle says that allele frequencies in a population are stable and are constant generation to generation. $p^2 + 2pq + q^2 = 1$	Hardy Weinberg principle
34.	In the given diagrammatic representation of operation of natural selection, identify a, b and c 	(a) Stabilising (b) Directional (c) Disruptive
35.	Natural selection can lead to stabilisation, directional change or disruption. True/False	True
36.	The first cellular forms of life appeared on earth about _____million years ago (mya)	2000
37.	Which of the following is the correct sequence of ancestor of the evolution of plant forms. a-Chlorophyte, b-Psilophyton, c-Rhynia type, d-Tracheophyte	a → d → c → b
38.	Jawless fish probably evolved around 500 mya . True/False	False (350 mya)
39.	In 1938, a fish caugnt in _____ happened to be a coelacanth which was thought to be extinct.	South Africa

40.	_____ evolved into the first amphibians.	Lobefins
41.	_____ were present but they all fell to form coal deposits slowly.	Giant ferns (Pteridophytes)
42.	About 200 mya the dinosaurs suddenly disappeared from the earth. (True/False)	False (It is 65 mya)
43.	*Dryopithecus* and *Ramapithecus* were existing about_____.	15 mya
44.	Two mya, which man ancestor probably lived in East African grasslands?	Australopithecines
45.	*Homo habilis* has the brain capacities about.	650-800 c.c
46.	*Homo erectus* had a large brain around 900 c.c and probably ate meat. True/False	True
47.	Who with a brain size of 1400 cc lived in near east and central Asia between 1,00000–40,000 years back.	Neanderthal man
48.	*Homo sapiens* arose in which place and moved across continents and developed into distinct races.	Africa
49.	The story of evolution of modern man is most interesting & appear to parellel evolution of human brain and language. True/ False	True
50.	Match the column	
	a. Ice age — i. East Aftrican grasslands	a. v
	b. Pre historic cave art — ii. *Ichthyosaurs*	b. iv
	c. *Australopithecus* — iii. *Tyrannosaurus*	c. i
	d. Fish like reptile — iv. 18000 yrs ago	d. ii
	e. Biggest dinosaurs — v. 75000–10000 yrs ago	e. iii
	f. *Homo erectus* — vi. Mammal ancestor	f. vii
	g. Therapsids — vii. 1.5 mya	g. vi
	h. Coelacanth — viii- Small and directional	h. x
	i. Darwinian variations — ix. random and direction less	i. viii
	j. Mutations — x. Lobefins	j. ix
51.	The correct sequence of human ancestors in increasing brain capacities. a. Neanderthal man b. *H. erectus* c. *H. habilis*	c, b, a

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\04-EVOLUTION.P65

HUMAN HEALTH AND DISEASE

Q.NO.	QUESTION	ANSWER
1.	In early times Hippocrates & Indian Ayurveda system asserted that state of body and mind are balanced by :-	Humors
2.	"Good Humor" hypothesis was given by :-	Hippocrates & Ayurveda
3.	"Good Humor" hypothesis was discarded by :-	William Harvey
4.	Health is affected by :-	Genetic disorder, infections & life style
5.	Health is maintained by :-	Balance diet, personal hygiene & regular exercise
6.	Health brings :-	Productivity, prosperity, longevity of people & decreased mortality of mother & infant
7.	Adversely affected organs characterised by signs and symptoms :-	Disease
8.	Disease which are transmitted from one person to another is known as :-	Infectious disease
9.	Major cause of death among non-infectious disease is :-	Cancer
10.	Disease causing agents are known as :-	Pathogens
11.	Due to interference of pathogen in normal vital activities results :-	Morphological & functional damage of cell
12.	For survival of pathogen within the environment of the host, pathogen have to :-	Adapt to life within the environment of the host.
13.	Transmission of "*Salmonella typhi*" through :-	Contaminated food & water
14.	Symptoms like sustained high fever (39° to 40°C), weakness, stomach pain, constipation, headache & loss of appetite are related with :-	Typhoid
15.	Death may occur in severe case of typhoid due to :-	Intestinal perforation
16.	Confirmation test for typhoid is :-	Widal
17.	Carrier of typhoid, who spread disease for several years was :-	Mary Mallon
18.	Bacteria responsible for pneumonia is :-	*Streptococcus pneumoniae* / *Haemophilus influenzae*
19.	Site of infection in pneumonia :-	Alveoli of lungs
20.	Symptoms like fever, chills, cough, headache and in severe case turning of color of nail from gray to bluish occurs in :-	Pneumonia
21.	Malaria is a _____ disease.	Protozoal
22.	The most serious type of malaria, malignant malaria which can even be fatal is caused by :-	*Plasmodium falciparum*
23.	Infectious form of *Plasmodium* for humans is :-	Sporozoites

24.	Transmitting agent for *Plasmodium* species is :-	Female *Anopheles* mosquito
25.	Toxin in malaria, which results in chill & high fever in humans is :-	Haemozoin
26.	The storage area for *Plasmodium* in female *Anopheles* mosquito is :-	Salivary gland
27.	Sexual stage which develops in RBCs is :-	Gametocytes
28.	Causative agent of a protozoan disease which cause amoebiasis is :-	*Entamoeba histolytica*
29.	Mechanical carrier for amoebic dysentery is :-	House fly
30.	Common round worm, which shows symptoms like internal bleeding, muscular pain, fever, anemia & blockage of intestinal passage is :-	*Ascaris*
31.	Filarial worm which cause chronic inflammation in lymphatic vessels, usually in lower limbs is :-	Elephantiasis or filariasis
32.	Genera which are responsible for fungal disease ringworm are :-	*Microsporum, Trichophyton* & *Epidermophyton*
33.	Dreadful disease which is completely eradicated is :-	Small pox
34.	Non specific type of defence that is present at the time of birth is :-	Innate immunity
35.	The main physical barrier of our body which prevents entry of micro-organism from environment is :-	Skin & mucosal epithelium
36.	Polymorpho-nuclear leucocytes (PMNL) which form cellular barrier are :-	Neutrophils
37.	Protein which is secreted by virus infected cell and form cytokine barrier is :-	Interferon
38.	When our body encounters first time with a pathogen. Then the first time low intensity response is known as :-	Primary response
39.	Highly intensified second time response in acquired immunity on the basis of previous memory is called as :-	Anamnestic response (secondary)
40.	Proteins which are produced by B-lymphocytes in response to pathogen are :-	Antibodies
41.	Formula which represent the antibody is :-	H_2L_2
42.	Antibody mediated immunity & cell mediated immunity are represented respectively by :-	B-lymphocyte & T-lymphocyte
43.	When the immune system of host are exposed to antigen and produces antibodies it is called as :-	Active immunity
44.	Yellowish fluid which contains IgA antibody and protects in initial day of infant is :-	Colostrum
45.	Principle of vaccination or immunisation is based on :-	Memory
46.	When preformed antibodies against the antigen are injected to patient and immunity is acquired than type of immunity is :-	Passive immunity
47.	Vaccine produced using method of recombinant DNA technology is :-	Hepatitis-B

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\05-HUMAN HEALTH AND DISEASE.P65

48.	Exaggerated response of immune system to certain antigens is known as :-	Allergy
49.	Due to genetic or unknown reasons, the immune system attacks on self cell, it is called as :-	Auto immunity
50.	Constitutes 50% of lymphoid tissue and located within the major respiratory, digestive & urogenital tract is known as :-	Mucosa associated lymphoid tissue (MALT)
51.	Enzyme which is required in AIDS disease to form viral DNA from viral RNA is :-	Reverse transcriptase
52.	Number of these cells decreases progressively during HIV infection leading to immunodeficiency.	Helper T-lymphocytes
53.	Infections occuring in AIDS due to immunodeficiency are caused by:-	*Mycobacterium*, viruses, fungi and parasites like *Toxoplasma*.
54.	Widely used diagnostic test for AIDS is :-	ELISA
55.	The cell which continues to produce virus and act like HIV factory is :-	Macrophage
56.	Phenomenon of cell by which it inhibits the uncontrolled growth is :	Contact inhibition
57.	The property of malignant tumor through which it spreads all over body is :-	Metastasis
58.	Physical, chemical or biological factors which transform a normal cell into cancerous cell, known as :-	Carcinogen
59.	Biological response modifier which activate immune system and help in destroying the tumor is :-	α-interferon
60.	White, odourless, bitter crystalline compound which are chemically diacetyl morphine is :-	Smack/Heroin
61.	Receptor for the opioid drugs are present in our body at :-	Central Nervous system and gastrointestinal tract
62.	Marijuana, hashish, charas and ganja are examles of :-	Cannabinoids
63.	Native to South America which interfere the transport of dopamine neurotransmitter is :-	Coca alkaloid or cocaine
64.	The first use of drug may be out of :-	Curiosity or experiment
65.	Anxiety, shakiness, nausea and sweating are relieved when use is resumed are known as :-	Withdrawal syndrome
66.	Reckless behaviour, vandalism violence etc. are immediate adverse effect of :-	Drug & alcohol abuse

STRATEGIES FOR ENHANCEMENT IN FOOD PRODUCTION

Q.NO.	QUESTION	ANSWER
1.	_____is the agricultural practice of breeding and raising livestock	Animal husbandary
2.	How much percent of the world livestock population is in India and China?	70%
3.	Name the branch of biology deals with rearing, catching, selling of fish, molluscs and crustacean.	Fishery
4.	_____ is the management of animals for milk and its products for human consumption.	Dairing
5.	In _____ we deals with the processes and systems that increases yield and improve quality of milk.	Dairy farm management
6.	_____is primarily dependent on the quality of breeds in the farm.	Milk yield
7.	_____ is the class of domesticated fowl used for food or for their eggs.	Poultry
8.	What is the aim of animal breeding?	Increasing the yield of animals and improving the desirable qualities of the produce
9.	What do we understand by the term breed?	A group of animals related by descent and similar in most characters like general appearance, features, size, configuration etc are said to belong to a breed.
10.	When breeding is between animals of more closely related individuals within the same breed it is called _____.	inbreeding
11.	When breeding is between animals of different breeds, it is called ____.	cross breeding
12.	Inbreeding refers to the mating of more closely related individuals within the same breed for _____ generations.	4 to 6
13.	The breeding strategy is as follows--_____males and _____females of the ____breed are identified and mated in pairs.	Superior , superior , same
14.	Inbreeding increases _____.	homozygosity
15.	What is inbreeding depression?	Continued inbreeding, especially close inbreeding, usually reduces fertility and even productivity
16.	How can we restore fertility and yield in case of inbreeding depression?	Selected animals of the breeding population should be mated with unrelated superior animals of the same breed.

17.	Outbreeding is the breeding of _____ animals which may be between individuals of the same breed but having_____	Unrelated , no common ancestors or between different breeds or different species
18.	What is cross breeding?	In this method , superior males of one breed are mated with superior females of another breed
19.	What is the significance of cross breeding?	It allows the desirable qualities of two different breeds to be continued The progeny hybrid animals may themselves be used for commercial production
20.	In this method , male and female animals of two different related species are mated.	Interspecific hybridization
21.	Controlled breeding experiments are carried out using _____.	artificial insemination
22.	_____ helps us overcome several problems of normal mating	artificial insemination
23.	The semen is collected from the male that is choosen as a parent and injected into the reproductive tract of the selected female by the breeder . This technique is _____.	artificial insemination
24.	In this method , a cow is administrated hormones with FSH like activity , to induce follicular maturation and super ovulation instead of one egg. This method is ___	Multiple Ovulation Embryo Transfer Technology (MOET)
25.	What is the significance of MOET?	High milk yielding breeds of females and high quality meat yielding bulls have been bred successfully to increase herd size in a short time
26.	_____ is the maintenance of hives of honey bees for the production of honey.	Bee keeping or apiculture
27.	Most common species of honey bee in India is	*Apis indica*
28.	Name some very common fresh water fishes that are eaten today	*Catla* , Rohu , Common carp
29.	Name some very common marine water fishes that are eaten today	*Hilsa*, Sardines, Mackerel, Pomfrets
30.	_____was dependent to a large extent on plant breeding techniques for development of high yielding and disease resistant varieties of wheat , rice , maize etc.	Green revolution
31.	_____is the purposeful manipulation of plant species in order to create desired plant types that are better suited for cultivation, giving better yields and are disease resistant.	Plant breeding
32.	Present day many crops are the result of _____	Domestication in ancient times
33.	_____ involves crossing or hybridization of _____ , followed by _____to produce plants with _____of higher yield , nutrition and _____to diseases	Classical plant breeding, pure lines, artificial selection, desirable traits, resistance

34.	If we were to list the traits or characters that the breeders have tried to incorporate into crop plants , the first we would list would be increased _____and improved____	Crop yield , quality
35.	_____ is the root of any breeding programme.	Genetic variability
36.	What do you meant by germplasm collection?	The entire collection having all the diverse alleles for all genes in a given crop is called germplasm collection
37.	What are the main steps in breeding a new genetic variety of a crop?	(a) collection of variability (b) evaluation and selection of parents (c) cross hybridisation among the selected parents (d) selection and testing of superior recombinants (e) testing, release and commercialization of new cultivars
38.	India is mainly an agricultural country . Agriculture accounts for approximately _____ per cent of India's GDP and employs nearly _____ per cent of the population.	33, 62
39.	During the period 1960 to 2000 , wheat production increased from _____ to _____whereas rice production went up from _____ to _____	11 million tones, 75 million tones, 35 million tones, 89.5 million tones
40.	Name the scientist who first of all develop semidwarf variety of wheat	Norman E. Borlaug
41.	In 1963 , several varieties such as _____ and _____, which were high yielding and disease resistant were introduced all over the wheat growing belt of India.	Sonalika, kalyan sona
42.	_____is the variety of sugarcane was originally grown in north india.	*Saccharum barberi*
43.	Tropical canes grown in south india _____ had thicker stems and higher sugar content but did not grow well in north india.	*Saccharum officinarum*
44.	Name some plant disease caused by fungi	Brown rust of wheat, red rot of sugarcane, late blight of potato
45.	Name some plant diseases caused by bacteria	Black rot of crucifers
46.	Name some plant diseases caused by viruses	Tobacco mosaic , Turnip mosaic
47.	The conventional method of breeding for disease resistance is that of ____ and ___	Hybridization , selection
48.	Complete the following table	

Crop	Variety	Resistant to diseases
Wheat	___a__	Leaf and stripe rust
__b____	Pusa swarnim	___c____
Cauliflower	Pusa Shubra	____d___
Cowpea	___e___	Bacterial blight
Chilli	Pusa sadabahar	____f____

(a)	Himgiri
(b)	Brassica
(c)	White rust
(d)	Black rot and Curl blight black rot
(e)	Pusa komal
(f)	Chilli mosaic virus, Tobacco mosaic virus, Leaf curl

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\06-STRATEGIES FOR ENHANCEMENT IN FOOD PRODUCTION.P65

49. _____ is the process by which genetic variations are created through changes in the base sequence within genes resulting in the creation of a new character or trait not found in parental type.

Mutation

50. In mung bean , resistance to _____ and _____were induced by mutations

Yellow mosaic virus, powdery mildew

51. Resistance to yellow mosaic virus in bhindi was transferred from a wild species and resulted in a new variety of Abelmoschus esculentus called _____

Parbhani kranti

52. Some released crop varieties bred by hybridization and selection, for insect pest resistance are given in the table, complete the table

Crop	Variety	Insect Pests
Brassica	___a____	Aphids
Flat bean	___b____	Jassids, __c___, and fruit borer
Okra	*Pusa sawani*	___d___

(a)	Pusa Gaurav
(b)	Pusa sem 2, Pusa sem 3
(c)	aphids
(d)	shoot and fruit borer

53. What is meant by biofortification ?

Breeding crops with higher levels of vitamins and minerals, or higher protein and healthier fats

54. Breeding for improved nutritional quality is undertaken with the objectives of improving the content and quality of _____, _____ wheaeras the content of _____,_____ and _____

Protein, oil, vitamin, micronutrient, mineral

55. In 2000 , maize hybrids that had twice the amount of the amino acids ____ and _____ , compared to existing maize hybrids were developed.

lysine, tryptophan

56. Write the full forms of the following
(a) SCP (b) IARI

(a) Single Cell Protein
(b) Indian Agricultural Research Institute

57. As traditional breeding techniques failed to keep pace with demand and to provide sufficiently fast and efficient systems for crop improvement , another technology called _____got developed

Tissue culture

58. What is totipotency?

Capacity to generate a whole plant from any cell / explants is called totipotency

59. Any part of a plant taken out and grown in a test tube, under sterile conditions in a special nutrient media that plant part is called

explant

60. The method of producing large number of plants in very short duration using plant explants by tissue culture is called

micropropagation

61. Plant produced by micropropagation will be genetically identical to the original plant from which they were grown ; they are called

Somaclones

62.	Isolated protoplasts from two different varieites of plants; can be fused to get hybrid protoplast. These hybrids are called	Somatic hybrids
63.	What are the objectives of animal breeding?	The purpose of animal breeding is to produce animals with high yield, faster growth and improved reproductive rate
64.	Name the methods employed in animal breeding	Inbreeding, Outbreeding, Cross breeding, interspecific hybridization, Artificial insemination, MOET
65.	Multiple ovulation embryo transfer is the best method employed in animal breeding. Why?	Because it results in the development of superior traits of both the breeds and the progeny produced are fertile and there is no inbreeding depression
66.	Which part of the plant is best suited for making virus free plants	The apical or axillary meristems
67.	What is the major advantage of producing plants by micropropagation?	It is faster method of producing a large number of plants. Plants formed by micropropagation are genetically identical to the original plant.
68.	Find out the various components of the medium used for propagation of an explants in vitro are ?	Sucrose (source for energy and carbon), inorganic salts, vitamins, growth regulators (auxins and cytokinins)
69.	Name any five hybrid varieties of crop plants which have been developed in India.	Sonalika (wheat), Himgiri (wheat), Pusa swarnim (mustard), Pusa sem 2 (bean), Pusa sawani (lady finger)

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\06-STRATEGIES FOR ENHANCEMENT IN FOOD PRODUCTION.P65

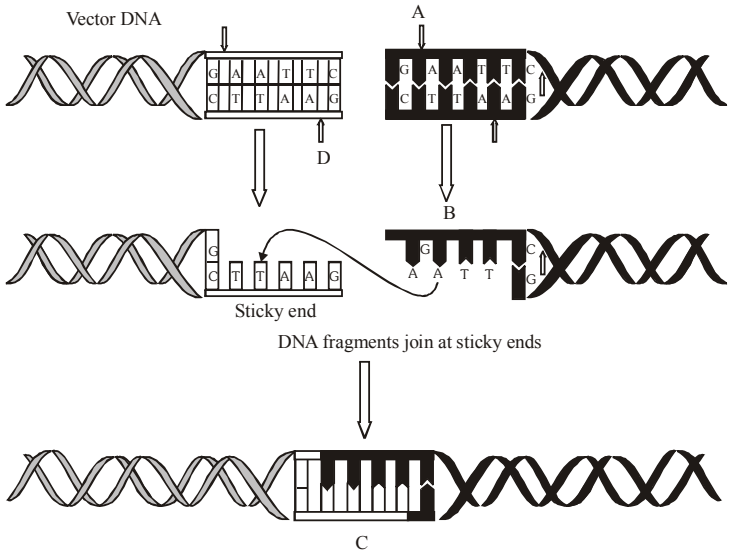
MICROBES IN HUMAN WELFARE

<table>
<tr><th>Q.NO.</th><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>1.</td><td>Which kingdoms of the living organisms contain microorganism?</td><td>Monera & Protista</td></tr>
<tr><td>2.</td><td>Which Microorganism grow in milk and convert it to curd ?</td><td>Lactic acid Bacteria (LAB)</td></tr>
<tr><td>3.</td><td>Which microorganism improves the nutritional quality of curd by increasing vitamin B_{12} ?</td><td>LAB</td></tr>
<tr><td>4.</td><td>The puffed-up appearance of dough is due to production of</td><td>CO_2 gas</td></tr>
<tr><td>5.</td><td>The dough which is used for making bread is fermented by which microbe ?</td><td>Saccharomyces cerevisiae (Baker's yeast)</td></tr>
<tr><td>6.</td><td>Which traditional drink in some part of southern India is made by fermenting sap of palms ?</td><td>Toddy</td></tr>
<tr><td>7.</td><td>Match the column
<table><tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr><tr><td>(1)</td><td>Propionibacterium Sharmanii</td><td>(a)</td><td>Penicillin</td></tr><tr><td>(2)</td><td>Saccharomyces cerevisiae</td><td>(b)</td><td>Citric acid</td></tr><tr><td>(3)</td><td>Aspergillus niger</td><td>(c)</td><td>Ethanol</td></tr><tr><td>(4)</td><td>Penicillium notatum</td><td>(d)</td><td>Swiss cheese</td></tr></table></td><td>1-d, 2,-c,3-b,4-a</td></tr>
<tr><td>8.</td><td>Production on an industrial scale, requires growing microbes in very large vessels called :-</td><td>Fermentor</td></tr>
<tr><td>9.</td><td>From given table choose that alcoholic drinks which are produced without distillation of fermented broth
Whisky, Wine
Brandy, Rum, Beer</td><td>Wine, Beer</td></tr>
<tr><td>10.</td><td>Name of the first antibiotic to be discovered :-</td><td>Penicillin</td></tr>
<tr><td>11.</td><td>Match the column
<table><tr><th></th><th>Column-A</th><th></th><th>Column-B</th></tr><tr><td>(1)</td><td>Acetobacter aceti</td><td>(a)</td><td>Cyclosporin A</td></tr><tr><td>(2)</td><td>Clostridium butylicum</td><td>(b)</td><td>Lactic acid</td></tr><tr><td>(3)</td><td>Lactobacillus</td><td>(c)</td><td>Butyric acid</td></tr><tr><td>(4)</td><td>Trichoderma Polysporum</td><td>(d)</td><td>Acetic acid</td></tr></table></td><td>1–d, 2–c, 3–b, 4–a</td></tr>
<tr><td>12.</td><td>Which group of enzymes are used in detergent formulation and are helpful in removing oily stains from the laundary</td><td>Lipases</td></tr>
</table>

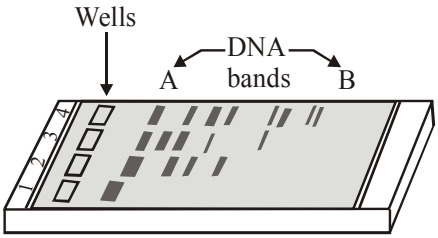
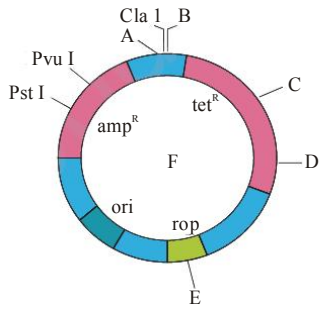
13.	Fill in the Blank (i) ……A….. is the blood-chloesterol lowering agent which is produced by the yeast…..B….. (ii) …..A…..Produced by the bacterium streptococcus and modified by genetic engineering is used as a…..B…..for removing….C… from blood vessel.	 A-Statin B-*Monascus purpureus* A-Streptokinase B-Clotbuster C-Clots
14.	Physical removal of large and small particles from the sewage through filtration and sedimentation is done under -	Primary treatment
15.	What is the flocs in Biological treatment step	Masses of bacteria associated with fungal filament to form mesh like structure
16.	The amount of oxygen that would be consumed if all organic matter in one liter of water oxidised by bacteria is called ?	BOD (Biochemical oxygen demand)
17.	Which of the following is True/False (i) The lower the BOD of waste water more is its polluting potential (ii) Secondary treatment of sewage waste also called Biological treatment (iii) Treatment of waste water is done by the Heterotrophic microbes naturally present in the sewage (iv) Sediment which treated with bacterial flocs is called activated sludge (v) A small part of activated sludge is pumped into large tank called anaerobic sludge digesters.	 False True True True False
18.	In anaerobic sludge digesters during digestion, bacteria produce mixture of gases called biogas. Give the three example of components of biogas.	(i) Methane (ii) Hydrogen sulphide (iii) Carbon dioxide
19.	Which two action plan initiated by the ministry of environment and forests to save major rivers of our country from pollution ?	Ganga action Plan & Yamuna action Plan
20.	The technology of biogas production was developed in India mainly due to the efforts of	IARI & KVIC
21.	Give the name of pathogen which attack on insect and other Arthopods so used as biological control agents.	Baculoviruses
22.	Majority of baculoviruses used as biological control agents belong to which genus ?	Nucleopolyhedrovirus
23.	The main sources of biofertilisers are….	Bacteria, fungi and cyanobacteria
24.	In Paddy fields which microorganism serve as important biofertiliser?	Cyanobacteria

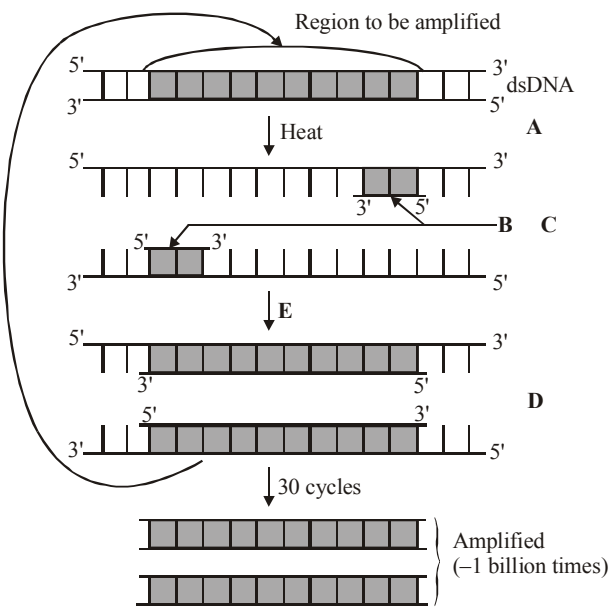
BIOTECHNOLOGY : PRINCIPLES AND PROCESSES

Q.NO.	QUESTION	ANSWER
1.	The techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans.	**Biotechnology**
2.	Who define biotechnology as the integration of natural science and organisms cells, parts thereof, and molecular analogues for products and services ?	**European Federation of Biotechnology (EFB)**
3.	Write down the two core techniques that enabled birth of modern biotechnology.	**(i) Genetic engineering (ii) Maintenance of sterile (microbial contamination-free) ambience and manufacture of biotechnological products.**
4.	Asexual reproduction preserves the _____, while sexual reproduction permits _____.	**Genetic information, variation**
5.	Which techniques of genetic engineering overcome the limitation and allows us to isolate and introduce only one or a set of desirable genes without introducing undersirable genes into the target organism ?	**Recombinant DNA, gene cloning and gene transfer**
6.	The fate of piece of DNA, which is somehow transferred into an alien organism would always integrated and multiply. (True/False)	**False**
7.	Name the specific sequence which is responsible for initiation of replication ?	**Origin of replication (ori.)**
8.	The technique which is responsible for making multiple identical copies of any template DNA called _____.	**Cloning**
9.	Autonomously replicating circular extra-chromosomal DNA in bacterial cell is called _____.	**Plasmid**
10.	Who constructed the first recombinant DNA ?	**Stanley Cohen and Herbert Boyer (1972)**
11.	Which enzyme is known as 'molecular scissors' ?	**Restriction endonuclease**
12.	The cutting of DNA at specific locations is become possible due to _____.	**Restriction endonuclease (Restriction enzyme)**
13.	When a cut piece of DNA linked with plasmid and trasfers the linked DNA then in this condition plasmid act as _____.	**Vector**
14.	Mosquito acts as an _____ to transfer _____ the into human body.	**Vector, malaria parasite**
15.	Which enzyme links the antibiotic resistance gene with the plasmid vector ?	**DNA ligase**
16.	When a cut piece of DNA linked with plasmid with the help of ligase enzyme then this result in formation of new DNA, known as	**Recombinant DNA** (r-DNA)

17.	What are the three basic steps in genetically modifying as organism?	**1. Identification of DNA with desirable genes** **2. Introduction of the identified DNA into the host** **3. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.**
18.	Name the enzyme that cut the DNA at specific site :-	**Restriction endonuclease**
19.	The first discovered restriction endonuclease enzyme was :-	***Hind II***
20.	Hind-II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs. This specific base sequence is known as the	**Recognition sequences**
21.	Today we know more than _____ that have been isolated from over _____ of bacteria each of which recognize difference recognition sequences.	**900 restriction enzymes, 230 strains**
22.	Match the column for enzyme EcoRI **Column A** **Column B** 1. E A. Order of enzyme isolation 2. Co B. genus 3. R C. name of strain 4. I D. species	**1–B, 2–D, 3–C, 4–A**
23.	Restriction enzymes belong to which class of enzymes ?	**Nucleases**
24.	Which nuclease remove nucleotides from the ends of the DNA ?	**Exonuclease**
25.	Which specific nucleotide sequences is recognizes by restriction endonuclease in DNA ?	**Palindromic nucleotide sequences.**
26.	What is the Recognition sequence of EcoRI ?	**5'GAATTC3'**
27.	Identify the A, B, C and D in given diagram. 	**A- Foreign DNA** **B- Sticky end** **C- Recombinant DNA** **D- EcoRI**
28.	The groups of letters that form the same words when read both forward and backward, eg. "MALAYALAM" is called _____.	**Palindrome sequence of base pairs**
29.	What is the process of insertion of recombinant DNA in bacterial cell ?	**Transformation**

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XI\ENGLISH\08-BIOTECHNOLOGY PRINCIPLES AND PROCESSES.P65

30.	The cell which contain recombinant DNA	**Transformed and recombinant cell**
31.	Name the technique which is use to separate the DNA fragments ?	**Gel electrophoresis**
32.	In electrophoresis which compound is use to stain DNA fragments	**Ethidium bromide**
33.	Identify the A and B in given diagram.  	**A- Largest, B-Smallest DNA fragment**
34.	Nowadays which is the most commonly used matrix for electrophoresis ?	**Agarose**
35.	From where the agarose extracted ?	**Sea weeds**
36.	Name the useful selectable markers for *E.coli* ?	**Ampicillin, chloramphenicol, tetracycline or kanamycin, resistance gene**
37.	Identify the A, B, C, D, E and F in the given diagram  	**A-EcoRI, B-Hind III, C-BamH I, D-Sal I, E-Pvu II and F-pBR322**
38.	The extraction of DNA fragment from agarose gel is called _____.	**Elution**
39.	Match the column **Column A** / **Column B** 1. Recombinant DNA technology — A. Vector 2. Cloning vehicle — B. Joining enzyme 3. Macromolecule separation — C. Gel electrophoresis 4. DNA ligase — D. Genetic engineering	**1-D, 2-A, 3-C, 4-B**
40.	Name the process which is use to check the progression of restriction enzyme digestion.	**Agarose gel electrophoresis**
41.	Which type of substrate produced colour for differentiation of recombinant and non-recombinant colonies ?	**Chromogenic substrate**
42.	True and false : 1. The uptake of DNA during trasformation is an active process. 2. In recombinant DNA, DNA inserted within the coding sequence of an enzyme beta galactosidase. 3. Bioreactor provides the optimal conditions for obtaining desired product. 4. In insertional inactivation process alien gene introduced in ori. sequence.	**True** **True** **True** **False**

43.	The process include separation and purification of desirable protein is called _____	**Downstream processing**
44.	After insertion inactivation of galactosidase gene which type of colonies do not produce any colour ?	**Recombinant colonies**
45.	The part of Ti Plasmid is transferred by Agrobactrium in dicot plant is called.	**T DNA**
46.	If any protein encoding gene is expressed in a heterologus host, is called a …… .	**Recombinant protein**
47.	Which type of ion use to increase the efficiency of uptake of DNA by bacterial cell ?	**Divalent cation (Ca^{+})**
48.	The bacterial cell which is able to uptake plasmid DNA is called _____.	**Competent cell**
49.	In biolistic or gene gun method DNA is coated with _____.	**Gold or Tungsten**
50.	By which method rDNA is directly injected into the nucleus of animal cell ?	**Microinjection**
51.	Which enzyme is use for removal of RNA molecule in DNA isolation process.	**Ribonuclease**
52.	In DNA isolation for the precipitation reaction which chemical is used?	**Chilled ethanol**
53.	Identify A, B, C, D, E in given diagram 	**A-Denaturation, B-Primers, C-Annealing, D-extension, E-Taq Polymerase and deoxynucleotides**
54.	From which bacteria thermo stable taq DNA polymerase is isolated ?	***Thermus aquaticus***

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XI\ENGLISH\08-BIOTECHNOLOGY PRINCIPLES AND PROCESSES.P65

BIOTECHNOLOGY AND ITS APPLICATION

Q.NO.	QUESTION	ANSWER
1.	Which technology is used to purify the protein / organic compound ?	Downstream processing technology
2.	Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called _____ .	Genetically modified organism (GMO)
3.	Which of the following statement is true / false (i) Genetic modification has made crops more tolerant to abiotic stress (ii) Genetic modification has increased reliance on chemical pesticides (iii) Agrochemical based agriculture is a option that can be thought for increasing food production (iv) The green revolution succeeded in tripling the food supply and it was enough to feed growing human population	 True False True False
4.	Which type of plants are created with the use of GM to supply alternative resources to industries ?	Tailor made plants
5.	Bt toxin is produced by a bacterium called _____.	Bacillus thuringiensis
6.	Which type of gene has been cloned from the bacteria and been expressed in the palnt to provide resistance to insects.	Bt toxin gene
7.	Name the orders of insects which killed by proteins that produce by some strains of Bacillus thuringiensis :-	Lepidopterans, Coleopterans and Dipterans
8.	Which type of pH is responsible for conversion of inactive Bt protoxin into active form of toxin	Alkaline pH
9.	Fill in the blanks : (i) The activated Bt toxin binds to the ___(a)____ and create ____(b)____ (ii) B.thuringiensis form a protein crystal during a particular ____(a)____ these crystals contain a toxic ____(b)____ (iii) The choice of Bt genes depends upon ____(a)____ and ____(b)____ as most Bt toxins are ____(c)____	 (a) Midgut epithelial cell (b) Pores (a) Phase of their growth (b) Insecticidal protein (a) The crop (b) the target pest (c) Insect group specific
10.	What is the name of that gene which is responsible for the coding of Bt toxin	cry gene
11.	Give the two name of cry gene which control the cotton bollworms :-	cry I Ac, cry II Ab
12.	Which nematode infects the roots of tobacco plants	Meloidegyne incognitia
13.	A novel strategy was adopted to prevent the infection of nematode was based on which process ?	RNA interference (RNAi)
14.	Which take place in all eukaryotic organisms as a method of cellular defense	RNAi
15.	In RNA the source of complementary RNA could be from an infection by viruses having RNA genomes or Mobile genetic elements (transposons) that form complementary RNA replicate via an _____ .	RNA intermediate

16.	Match the column : **Column-A** **Column-B** (1) RNAi (a) Corn borer (2) cry I Ab (b) Cotton bollworms (3) cry I Ac (c) Proinsulin (4) C peptide (d) Silencing of specific mRNA	1-d, 2-a, 3-b, 4-c
17.	In earlier insulin used for diabetes was extracted from pancreas of which animals?	Slaughtered cattle and pigs
18.	In mature insulin chain-A and chain-B are linked together by which bond?	Disulphide bridge
19.	Which peptide chain is removed during maturation of proinsulin	C-Peptide
20.	What is the main challenge for production of insulin using r-DNA techniques	Getting insulin assembled into a mature form.
21.	If a person is born with a hereditary disease, which corrective therapy be taken for such a disease	Gene therapy
22.	The first clinical gene therapy was given in _____ .	1990
23.	Adenosine deaminase enzyme is crucical for which system to function ?	Immune system
24.	Give the name of any three techniques that serve the purpose of early diagnosis	r-DNA technology, ELISA & PCR
25.	Very low concentration of a bacteria or virus (at a time when the symptoms of the diseae are not yet visible) can be detected by	Amplification of their nucleic acid by PCR
26.	Which technique is now routinely used to detect HIV in suspected AIDS patients?	PCR
27.	ELISA which is a molecular diagnosis techiques is based on the which principle?	Antigen antibody interaction
28.	Animal that have had their DNA manipulated to possess and express an extra (foreign) gene are known as :-	Transgenic animals
29.	Human protein α-1-antitrypsin used to treat _____ .	Emphysema
30.	Name of the first transgenic cow is :-	Rosie
31.	The milk of first transgenic cow contained a human protein ____ .	Human alpha-lactalbumin
32.	Which organisation in India make decisions regarding the validity of GM research and safety of introducing GM organisms for public services ?	GEAC (Genetic engineering approval committee)
33.	The use of bioresources of multinational companies and other organisation without proper authorisation from the countries and people concerned without compensatory payment is called :-	Biopiracy

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\09-BIOTECHNOLOGY AND ITS APPLICATION.P65

ORGANISMS AND POPULATIONS

<table>
<tr><th>Q.NO.</th><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>1.</td><td>Match column-I with column-II
<table>
<tr><th colspan="2">Column - I</th><th colspan="2">Column - II</th></tr>
<tr><td>1.</td><td>Commensalism</td><td>a.</td><td>+, –</td></tr>
<tr><td>2.</td><td>Amensalism</td><td>b.</td><td>0, –</td></tr>
<tr><td>3.</td><td>Competition</td><td>c.</td><td>0, +</td></tr>
<tr><td>4.</td><td>Predation</td><td>d.</td><td>–, –</td></tr>
<tr><td>5.</td><td>Mutualism</td><td>e.</td><td>+, +</td></tr>
</table></td><td>1-c, 2-b, 3-d, 4-a, 5-e</td></tr>
<tr><td>2.</td><td>Mutualism is an <u>x</u> whereas proto-co-operation is a <u>y</u> relation in which both interacting species are benefitted.</td><td>x = Obligatory
y = Non-obligatory</td></tr>
<tr><td>3.</td><td>Interaction in which either one or both participating species are harmed such interactions are called _____.</td><td>Detrimental or negative interactions</td></tr>
<tr><td>4.</td><td>Which organism shows the following properties in an ecosystem.
• Act as conduit of energy transfer
• Maintain species diversity in community
• Regulate the population of prey species
• Prudent in nature</td><td>Predators</td></tr>
<tr><td>5.</td><td>More than 10 species of invertebrates became extinct within a year from the American Pacific Coast when Pisaster starfish was removed. This extinction was due to -</td><td>Interspecific competition</td></tr>
<tr><td>6.</td><td>Some insects and frogs are cryptically coloured to avoid being detected by predators. This is called -</td><td>Camouflage</td></tr>
<tr><td>7.</td><td>How much percentage of all insects are phytophagous which feeds on plant sap and other plant part ?</td><td>≈ 25%</td></tr>
<tr><td>8.</td><td>Match the columns regarding defence mechanisms adopted by prey species to be protected from predators.
<table>
<tr><th colspan="2">Column - A</th><th colspan="2">Column - B</th></tr>
<tr><td>A.</td><td>Cardiac glycosides</td><td>i.</td><td>Acacia, Cactus</td></tr>
<tr><td>B.</td><td>Distasteful due to a special chemical</td><td>ii.</td><td>Insects</td></tr>
<tr><td>C.</td><td>Thorns</td><td>iii.</td><td>Calotropis</td></tr>
<tr><td>D.</td><td>Camouflage</td><td>iv.</td><td>Monarch butterfly</td></tr>
</table></td><td>A-(iii), B-(iv), C-(i), D-(ii)</td></tr>
<tr><td>9.</td><td>When a species is introduced into new geographical area becomes invasive and causes havoc by spreading rapidly, due to _____</td><td>Absence of its natural predator</td></tr>
<tr><td>10.</td><td>When feeding efficiency of one species is reduced in presence of the other species even if the resources are abundant, it is called -</td><td>Interference competition</td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

11. Match the following columns -

Column - I		Column - II	
1.	Resource partitioning	A.	Gause
2.	Competitive exclusion	B.	Darwin
3.	Struggle for existence	C.	Mac Arthur

1 - C, 2 - A, 3 - B

12. Which organisms are more adversely affected by competition than carnivores ?

Plants and Herbivores

13. To avoid competition species choose different feeding time and different foraging patterns this is called _____ .

Resource partitioning

14. How many of the following statements are *incorrect* for *parasites ?*
(A) Presence of un-neccessary sense organs
(B) Presence of adhesive organs & suckers
(C) Developed digestive system
(D) High biotic potential

Two (A and C are incorrect)

15. Human liver fluke is a _A_ which depends on two intermediate hosts _B_.

A = Trematode parasite
B = a snail and a fish

16. An organism reduces survival, growth and reproduction of other organism, reduces its population density and makes it physically weak, such organism is called _____.

Parasite

17. How many of the following organisms are considered as Ecto-parasites ?
Lice, Ticks, Copepods, *Cuscuta*, *Taenia*, *Ascaris*, Liver fluke

Four
(Lice, Ticks, Copepods, *Cuscuta*)

18. How many of the following are specific properties of endo-parasites ?
(A) More complex life cycle
(B) High reproductive potential
(C) Simple morphology
(D) Simplified anatomy

Four (A, B, C, D)

19. Interaction in which one organism lays its eggs in the nest of another organism is called _x_. For example _y_.

x = Brood parasitism
y = Cuckoo bird and crow

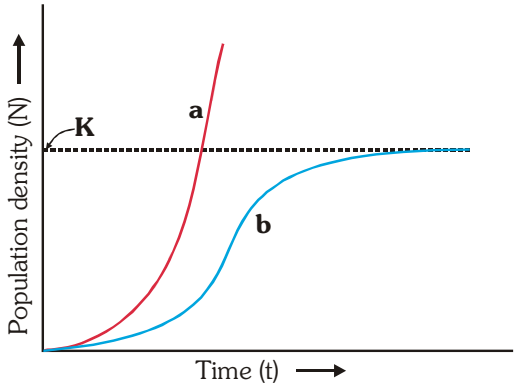
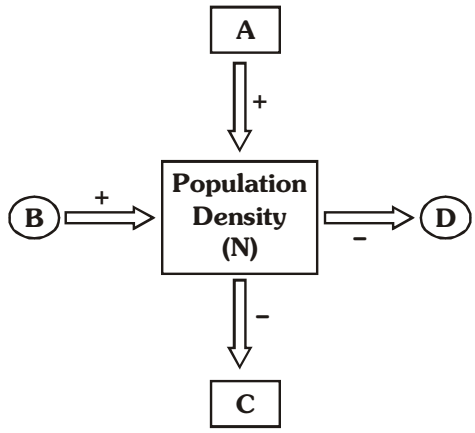
20. Match the following organisms according to their commensal relationship.

1.	Mango Branch	A.	Sea anemone
2.	Clown fish	B.	Barnacles
3.	Egret bird	C.	Cattle
4.	Whale	D.	Orchid

1 - D, 2 - A, 3 - C, 4 - B

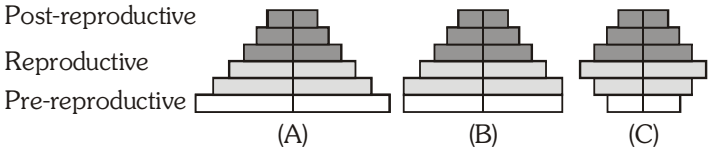
Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\10-ECOLOGY.P65

21.	The mutualistic relationship between fungus and cyanobacteria is called _____ .	Lichen
22.	In an interspecific interaction a filamentous eukaryotic organism absorbs nutrients from the soil for the higher plant. In turn the host provides some energy yielding carbohydrates, such association is called _____ .	Mycorrhiza
23.	Pollinators and seed dispersers are offered reward by plants in form of _A_ and _B_ respectively.	A = Pollen & nectar B = Juicy and nutritious fruits
24.	In which of the following interaction both interacting species show co-evolution, co-extinction, co-existence ? (1) Competition (2) Predation (3) Mutualism (4) Commensalism	Mutualism
25.	In symbiotic relationship between fig tree and wasp, the wasp uses fig fruit for _x_ and developing seeds for _y_. In turn favours _z_.	x = Oviposition (egg laying) y = Nourishing its larva z = Pollination of fig.
26.	Select the correct option with respect to the related organisms and interactions. **Interactions** **Related organism** A. Mutualism 1. Sparrow and seed B. Predation 2. Goat and Abingdon tortoise C. Parasitism 3. Flamingoes and fishes D. Competition 4. Monarch butterfly and bird E. Commensalism 5. Hedge plant and *Cuscuta* 6. Orchid and mango tree 7. Human and *Ascaris* 8. Orchid and Bee 9. *Balanus* and *Chathamalus*	A – 8 B – 1, 4 C – 5, 7 D – 2, 3, 9 E – 6
27.	True or False 1) 99% of animals and nearly all plants are conformers. 2) Thermoregulation is energetically expensive for many organisms. 3) Heat loss or heat gain is a function of surface area. 4) All birds and mammals are regulators.	 True True True True
28.	Under unfavourable conditions, a stage of suspended development is called _____.	Diapause
29.	Many tribes live in the high altitude of Himalaya, have _____ red blood cell count than people living in the plains.	Higher
30.	Symptoms of altitude sickness are _____.	Nausea, Fatigue, Heart palpitations
31.	How does human body compensate to altitude sickness ?	Increasing RBC production, Increasing breathing rate, Decreasing binding affinity of Hb

32.	Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. This is called _____.	Allen's rule
33.	How does kangaroo rat fulfill its water requirements ?	By internal fat oxidation
34.	How do Antarctic fish manage to keep their body fluids from freezing ?	They have antifreeze proteins
35.	Under normal conditions, _____ and _____ are the most important factors influencing population density.	Births, Deaths
36.	Exponential growth of population is seen when resources are ____	Unlimited
37.	Exponential growth equation is _____	$N_t = N_0 e^{rt}$ or $\frac{dN}{dt} = rN$
38.	Logistic growth equation is _____	$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
39.	Write some examples of organisms that breed only once in their lifetime.	Pacific salmon fish and Bamboo plant
40.	Write equations for curves a and b in the graph given below. 	$a = \frac{dN}{dt} = rN$ $b = \frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
41.	Identify A, B, C & D 	A – Immigration B – Natality C – Emigration D – Mortality

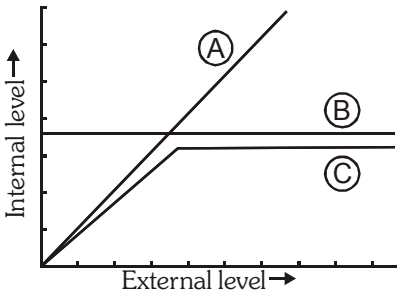
Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\10-ECOLOGY.P65

42. Identify the growth status of the population from given age pyramids :-



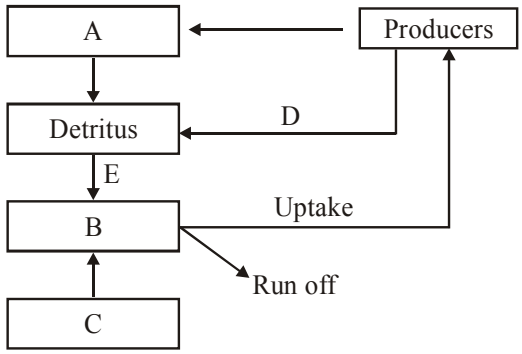
A – Expanding
B – Stable
C – Declining

43. Identify A, B & C –



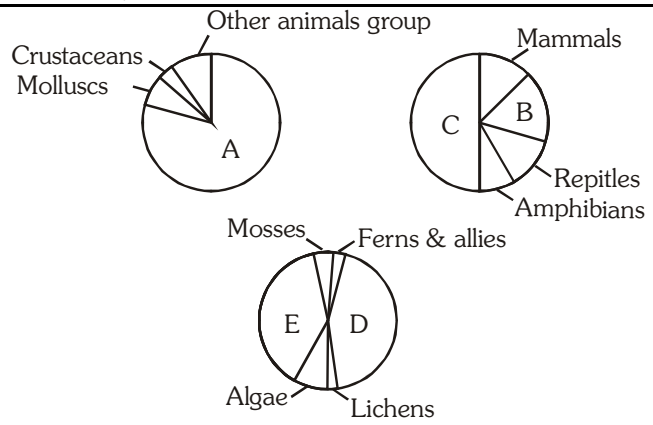
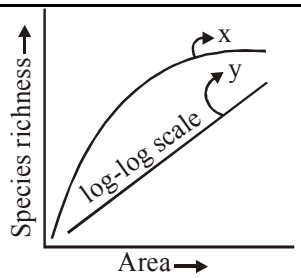
A – Conformers
B – Regulators
C – Partial regulators

ECOSYSTEM

Q.NO.	QUESTION	ANSWER
1.	How many of the following can be categorised under man-made ecosystem ? Lake, Wetland, Pond, Cropland, River, Aquarium, Estuary	Cropland and Aquarium (Pond may be natural or man made)
2.	Different species of plants occupy various levels in the forest ecosystem. Such distribution is called _____.	Stratification or vertical stratification
3.	The basic requirement for any ecosystem to function and sustain is	Constant solar input
4.	Production and productivity can be represented by _x_ and _y_ respectively.	x = gm/m^2 y = $gm/m^2/yr$
5.	Representation of production and productivity in terms of energy__	$kcal/m^2$(Production) $kcal/m^2/yr$(Productivity)
6.	A rabbit feeds on grass and gains 1 kg weight within two months, this rate of biomass gain in termed as _____.	Secondary productivity.
7.	The NPP of whole biosphere is _A_ out of which NPP of oceans is _B_.	A = 170 billion tons B = 55 billion tons
8.	Despite covering 70% of surface, oceans show very low productivity because _____.	Producers are micro-organisms (Phytoplankton), N_2 deficiency, light limiting factor in aquatic ecosystem
9.	 Identify A, B, C, D and E in above cycle.	A = Consumer B = Soil solution C = Rock minerals D = Litter fall E = Decomposition
10.	How much amount of carbon is fixed in biosphere through photosynthesis annually?	4×10^{13} kg
11.	The price tag given to fundamental ecosystem services is approximately _____ .	US $33 trillion
12.	If 1,000,000 J of sunlight is provided to a food chain what is the amount of energy is present at top consumer level if food chain have four trophic level.	10 J (Because only 1% of total sunlight is available to plants)

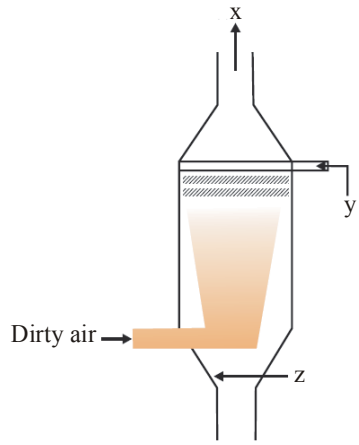
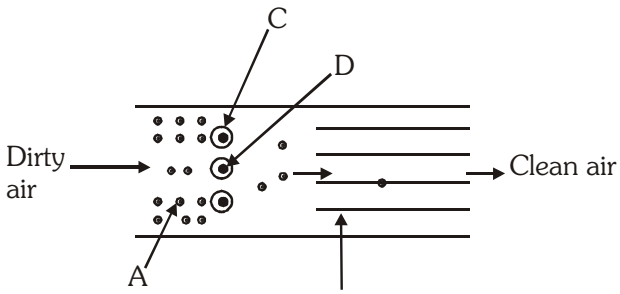
Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\10-ECOLOGY.P65

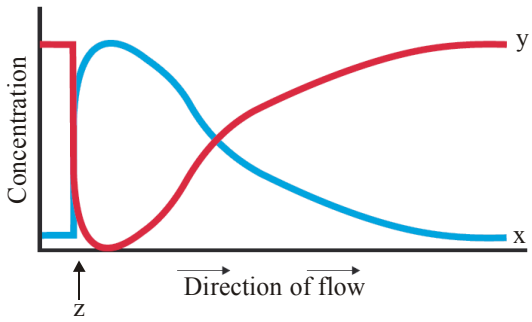
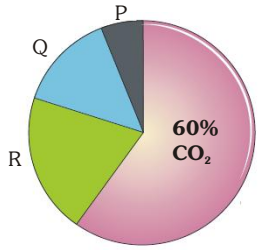
BIODIVERSITY AND CONSERVATION

Q.NO.	QUESTION	ANSWER
1.	Term biodiversity was popularised by -	Edward Wilson
2.	Biodiversity is the combined diversity at all levels of biological organisation what are those levels ?	(i) Genetic diversity (ii) Species diversity (iii) Ecological diversity
3.	How many species of plants and animals are described according to IUCN (2004) ?	> 1.5 million
4.	How much percentage of recorded species are animals and plants respectively?	Animals - > 70% Plants - 22%
5.	 Study the above pie charts and match the following (i) Angiosperms (ii) Fishes (iii) Birds (iv) Fungi (v) Insects	A-Insects (v) B-Birds (iii) C-Fishes (ii) D-Angiosperm (i) E-Fungi (iv)
6.	India has only _x_% of world's land area but have _y_% of global species diversity.	x = 2.4% y = 8.1%
7.	The species diveristy _____ when we move towards the equator whereas it _____ when we move towards the poles.	Increases Decreases
8.	 Study the above graph and answer the following questions - (i) Identify x and y (ii) The relation between species richness and area can be denoted as generally (iii) The value of Z lies in range of ____, regardless of the taxonomic group or the region (whether it is plants in Britain, birds in California or molluscs in New York state.) (iv) On a log scale the relation between species richness and area is	(i) x = ($S = CA^z$), y = (log S = log C + Z log A) (ii) Rectangular hyperbola (iii) Z = 0.1 to 0.2 (iv) Straight line

9.	According to rivet popper hypothesis, if ecosystem is considered as airplane then what does the rivets on wings represent.	Key stone species
10.	Match the following organism according to their place of extinctions :- A. Dodo — i. Australia B. Steller's sea cow — ii. Mauritius C. Quagga — iii. Africa D. Thylacine — iv. Russia	A(ii), B(iv), C(iii), D(i)
11.	Loss of biodiveristy in a region may result in (A) Increased variability in plant productivity, water use, pests and disease cycle. (B) Decline in plant production (C) Lowered resistance to environmental perturbations How many of the above statements are correct ?	A, B, C (All)
12.	Select the terms which are related to loss of biodiveristy and considered under The Evil Quartet - Co-extinction, Proto-cooperation, Over exploitation, Predation, Endemism, Alien - species invasion, Habitat loss and fragmentation.	Four (Co-extinction, Over exploitation, Alien - species invasion, Habitat loss and fragmentation)
13.	Which area on earth is also called lungs of planet ? Why ?	Amazon rains forest (It produces 20% of total O_2 in earth atmosphere)
14.	Select the species which are extincted due to over exploitation :- Nile perch, Water hyacinth, African catfish, Passenger pigeon, *Lantana*, Cichlid fish, Steller's sea cow, Quagga, Dodo.	Two (Passenger pigeon, Steller's sea cow)
15.	Exploring molecular, genetic, species - level diversity for products of economic importance is termed as	Bioprospecting
16.	Biodiversity conservation can be grouped into how many categories?	(1) Narrowly utilitatian (2) Broadly utilitatian (3) Ethical

ENVIRONMENTAL ISSUES

Q.NO.	QUESTION	ANSWER
1.	Which act was passed by Govt. of India to control environmental pollution, to protect and improve the quality of air, water and soil?	Environmental protection act(1986)
2.	Observe the above figure and answer the following questions - (i) Identify the above figure. (ii) Mention x, y and z. (iii) Above setup is used to remove _____. 	(i) Scrubber (ii) x = Clean air y = water/lime spray z = Particulate matter (iii) Particulate matter and gases like SO_2.
3.	 (i) Identify the above figure. (ii) Mention its efficiency (iii) Label A, B, C, D (iv) Air velocity in exhaust should be_____ and electrode wires should be maintained at _____.	(i) ESP (Electrostatic precipitator) (ii) Removes 99% of particulate matter (iii) A = Dust particle B = Collection plate grounded C = Discharge corona D = Negatively charged wire (iv) Low ; several thousand volts
4.	Full form of CPCB ?	Central Pollution Control Board
5.	According to CPCB which size particles are responsible for causing the greateest harm health ?	2.5 µm (PM 2.5) or less in diameter (PM 2.5)
6.	Metals used in catalytic converters are -	Platinum, Palladium, Rhodium
7.	Which act was amended to include noise pollution as air polution ?	Air prevention and control of pollution act (1981) was amended in 1987.

8.	 Observe the above graph and identify x, y and z.	x = BOD, y = Dissolved oxygen z = Sewage discharge
9.	Which aquatic plant is also called world's most problematic aquatic weed or Terror of Bengal ?	Water hyacinth (*Eichhornia crassipes*)
10.	Increase in concentration of toxic substances at successive trophic levels due to their accumulations, it is called -	Biomagnification
11.	Water→Zooplankton→Small fish→Large fish→Fish eating birds In the above food chains if concentrations of DDT is 0.003 ppb in water, then mention its concentration at successive trophic levels.	Water (0.003 ppb) ↓ ZP (0.04 ppm) ↓ SF (0.5 ppm) ↓ LF (2 ppm) ↓ Birds (25 ppm)
12.	Define eutrophication ?	Natural ageing of lake by nutrient enrichment of its water
13.	Cultural or accelerated eutrophication is -	Human induced eutrophication.
14.	Full forms of FOAM	Friends of Arcata Marshes
15.	In which parts we can see working ecosan toilets ?	Sri-Lanka and Kerala
16.	Identify P, Q, R in the composition of green house gases represented below – 	P = N_2O – 6% Q = CFCs – 14% R = CH_4 – 20%
17.	Good ozone and Bad ozone are found is _____ and _____ respectively.	Stratosphere and troposphere
18.	Unit used to measure thickness of ozone layer is stratosphere is–	Dobson unit (DU)

Z:\NODE02\B0B0-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\10-ECOLOGY.P65

19.	Which international treaty was signed to control the emission of ODS (Ozone depleting substances)?	Montreal Protocol (1987) became effective in 1989
20.	Match the following– (1) Chipko movement — (A) 1987 (2) Joint forest management — (B) 1980s (3) Air (prevention and control of pollution) act — (C) 1974 (4) Montreal protocol — (D) 1981	(1)-(C), (2)-(B), (3)-(D), (4)-(A)

Z:\NODE02\B080-BA\NCR_QUESTIONS_BIO\CLASS_XII\ENGLISH\10-ECOLOGY.P65